WATER RESOURCES, WELLS & PUMPS · NOTE PU04

Turbine and Special Pumps — Jet, Vertical Turbine, Submersible, Propeller and Airlift Pumps

Course topics: IDE-511 T5, IDE-511 T4

Jet pumps

- A centrifugal pump plus a jet (ejector). The centrifugal pump and its motor or engine sit at the surface or on the well and supply the driving head; the **jet assembly — nozzle and venturi** — sits in the well below the water.
- **Working:** part of the high-pressure delivery returns down a pressure pipe to the nozzle, which narrows the flow and speeds it up; the fast jet creates low pressure round the venturi and draws in more well water; the venturi widens gradually to the suction-pipe size, slowing the flow with little turbulence; the impeller's vacuum draws the water up and delivers it at the desired pressure. A central valve, suitably adjusted, keeps the pressure and flow while the driving water recirculates.
- **Efficiency:** low — at most about 35 per cent.
- **Use:** where the moving parts and prime mover must stay at the surface; fits tube wells as small as 5 cm inside diameter (a single-pipe ejector where two pipes will not fit).
- **Advantage:** high suction lift — 12 to 18 m with low- to medium-size jet pumps, against 6.5 m for a volute centrifugal pump.

Vertical turbine (deep-well turbine) pumps

- A vertical-shaft, diffuser-type, single- or multi-stage centrifugal pump whose pumping element hangs from the discharge pipe.
- Used in tube wells and open wells where the water is below the practical suction limit of centrifugal pumps; efficiency comparable with a good horizontal centrifugal pump; long, dependable service — but costlier and harder to inspect and repair.
- **Working:** water thrown from the impeller is turned upward by stationary guide vanes, and its kinetic energy becomes pressure in the casing. For high lifts, impellers are stacked in series in bowls, each sending water up into the eye of the next. Set the bowls deep enough to keep the impellers submerged at full capacity.

Part	Details
Pump head	Vertical driving motor and discharge connection
Pump unit (below ground)	Pump column, shafting and stages; a screen at the bottom keeps out coarse sand and gravel; always below the pumping water level
Shaft and column	Stainless-steel drive shaft in the centre of the column, surrounded by discharge water, in 3 m lengths joined by threaded couplings, with non-corrosive sleeves at the bearings; runs in resilient line-shaft bearings held in bearing couplings between 3 m column sections
Water lubrication	Rubber bearings, lubricated only when wet — after a long idle period pour water down the shaft before starting
Oil lubrication	Bearings in a small cover pipe with bronze bearings centring the shaft; keep the oil reservoirs filled (automatic oilers)
Impellers and stages	Impeller size is limited by the bowl, and the bowl by the well diameter; one stage gives little head, so several are usual — total head ∝ number of stages. Closed impellers are more efficient; semi-open ones resist sand locking

- **Curves** depend on bowl design and shaft speed and resemble centrifugal-pump curves, but turbine pumps are more sensitive to speed changes — study them carefully.
- **Selection:** prefer oil lubrication for wells producing fine sand or clay, water lubrication for domestic supply (oil-lubricated pumps lose some oil into the water). Test the well first and match the pump to its depth, inside diameter, static level, seasonal fluctuation and drawdown–yield curve. A poor match brings hydraulic and mechanical trouble, over-pumping, cavitation, vibration and wear.
- **Installation:** align pump and power unit; keep the pump clear of the casing (contact causes vibration and wears holes in it). Use a concrete foundation with at least 30 cm of bearing all round (measured from the outside of the gravel pack in gravel-packed wells, with two openings for refilling gravel). In an open dug well, hang the pump freely from two steel beams on masonry or concrete, with the strainer at least 1.5 m above the bottom to avoid sucking sand.
- **Drives:** electric motor, or an engine through a right-angle gear drive (costly but efficient) or a pulley head; run at rated speed.

Turbine pump trouble	Causes
Won't start	Faulty wiring or motor; bearing friction from misalignment, poor or wrong lubrication or dry rubber bearings; wrong impeller adjustment; impellers locked by wood, rags, metal, sand, corrosion or graphitisation
No water	Bowls not submerged (not primed); wrong rotation; head too high; over-pumping; speed too low; suction clogged; broken shaft, loose impellers, leaking column joints
Capacity too low	Head too high; speed too low; poor suction (turbulence, vortices, air leaks, vacuum from an inadequate vent); leaking threads, flanges or gaskets; wrong impeller adjustment; worn impellers
Too much power	Off-design conditions; wrong rotation; speed too high; different liquid; wrong impeller adjustment or lubrication; packing too tight; mechanical failure
Vibrates	Misalignment; bearing failure; poor suction; imbalance from foreign matter; worn rotating parts
Water in the shaft enclosing tube	Enclosing tube or bowl failure

Submersible pumps

- A turbine pump close-coupled to a submersible electric motor; pump and motor always run under the well water. Components: pump-and-motor assembly, head assembly, discharge column and power cable.
- A very short stainless-steel shaft carries impellers in series — closed, semi-open or open (closed for high pressure); water enters through a screen between motor and pump.
- The compact unit hangs from the discharge pipe; the only links to the surface are a submarine cable and a small copper oil tube.
- **Motor:** slim and long to fit small tube wells; usually the "dry" type, filled with non-hygroscopic mineral oil of high dielectric strength circulating through the windings; oil topped up through the copper tube (some have run for years without change); a mercury seal in a rotating cup on the shaft keeps water out at the top.
- **Service:** built for long unattended running, but if the motor fails the whole unit must be pulled; cable damage is the common fault — ensure reliable cable and prompt replacement before choosing one.
- **Suitability:** tube wells of 10 cm diameter or more; deep and crooked wells (short shaft); no working parts above ground; usable in flood-prone areas if the well is sealed and the starter and electrics sit above high-water level.
- **Installation:** cheap and easy — only the length of discharge pipe needed to keep the unit always submerged; rest the head assembly on a concrete base and fasten it; no pump house, but keep the starter and electrics in a waterproof box.

Propeller pumps (axial and mixed flow)

- For low heads and large capacities; flow through the impeller is parallel to the shaft, not radial. Main parts like a deep-well turbine: head, discharge column, vertical drive shaft and impeller; single- or multi-stage. The two types differ in their impellers.
- **Axial flow (screw type):** a boat-propeller impeller lifts the water by blade lift without changing its direction. Single stage: heads up to about 3 m; with more stages, 9 to 12 m.

- **Mixed flow:** open-vane, curved-blade impeller combining screw and centrifugal action; for large capacities at moderate heads; small single-stage pumps work efficiently at 2 to 3 m, larger and multi-stage ones at much higher heads; more efficient than axial flow at higher heads.
- **Pros:** low cost; more water than a centrifugal pump of the same impeller size. **Cons:** low lift limits them to canals, rivers, large shallow high-yield wells and drainage pumping; unsuitable where delivery must be throttled.
- **Characteristics:** flatter efficiency curve and much lower heads than centrifugal pumps; power falls continuously as discharge rises. No suction lift — submerge the bowl as the maker specifies.

Airlift pumps

- Work on the difference in specific weight: compressed air injected into a deeply submerged rising main lightens the air–water column inside it, so the heavier surrounding water pushes the mixture up. How high it rises depends on the submergence and air pressure.
- **Plant** (general arrangement as in Fig. 6.48 of the textbook): air compressor (single-stage below 4 kg per sq cm, two-stage above), air receiver, air pipe and rising main.
- **Pros:** simple; the tube well need not be straight or vertical; dirty water does no harm. **Cons:** low efficiency (average about 30 per cent); needs a deep well for submergence; high first cost including the compressor.
- **Improve it:** run the air pipe outside the rising main where space allows (inside, friction cuts output); flare the bottom of the rising main so water starts gently (much less power); give the delivery end a long gradual taper; direct the air upward.
- **Uses:** testing and cleaning tube wells; crooked wells where turbine pumps will not work.

Sources: ICAR eCourse 'Groundwater, Wells and Pumps' — Dr M. K. Jha (IIT Kharagpur), reviewed by Dr P. B. S. Sarma (IARI New Delhi). Lessons on pumps (classification, centrifugal and turbine pumps, pumping power). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. The main advantage of a jet pump is

(a) high efficiency (b) high suction lift
(c) low cost (d) no priming

Q2. Maximum efficiency of a jet pump is about

(a) 35% (b) 60%
(c) 75% (d) 90%

Q3. A submersible pump is a turbine pump close-coupled to

(a) a diesel engine (b) a submersible electric motor
(c) a windmill (d) a tractor PTO

Q4. Submersible pumps suit tube wells of diameter

(a) 5 cm or more (b) 10 cm or more
(c) 30 cm or more (d) 60 cm or more

Q5. A single-stage axial-flow propeller pump is limited to heads of about

(a) 3 m (b) 12 m
(c) 30 m (d) 60 m

Q6. Average efficiency of an airlift pump is about

(a) 10% (b) 30%

(c) 60% (d) 85%

Answers

Q1 — (b) high suction lift — *12–18 m vs 6.5 m for a volute pump.*

Q2 — (a) 35%

Q3 — (b) a submersible electric motor

Q4 — (b) 10 cm or more

Q5 — (a) 3 m

Q6 — (b) 30%
